UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5054 PHYSICS

5054/41 Paper 4 (Alternative to Practical), maximum raw mark 30

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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1 (a) (i) (*h* marked on) vertical line from floor to **bottom** of pendulum — B1 [1]

(ii) eye / E labelled level with bottom of pendulum/candidate's line in **(a)(i)** — B1 [1]

(b) (i) time too large / lose count / amplitude decays / stops swinging — B1 [1]

(ii)

T	T^2
3.062 / 3.06	9.38 / 9.36
2.992 / 2.99	8.95 / 8.94
2.918 / 2.92	8.51 / 8.53
2.846 / 2.85	8.10 / 8.12
2.774 / 2.77	7.70 / 7.67
2.698 / 2.70	7.28 / 7.29

T values correct to 3 or 4 s.f. — B1
T^2 values correct (ecf T) — B1
T^2 to 3 sf — B1 [3]

(iii) axes: correct way round, labelled quantity and unit — B1
scales: more than ½ grid, sensible — B1
points plotted accurately within ½ small square — B1
best fit straight line neatly drawn — B1 [4]

(iv) straight line / linear **AND** negative gradient / inversely related / one increases other decreases — B1 [1]

[Total: 11]

2 (a) goes down — B1 [1]

(b) use a **known mass** (stated or implied) with link to cm / marks on scale — B1 [1]

(c) (i) 1.5 cao — B1 [1]

(ii) 3.8 cao — B1 [1]

(iii) 0.76 (unit not required) ecf 0.3 × **(c)(ii)** / **(c)(i)** — B1 [1]

(d) no electricity needed / cheaper / easy to build / more sensitive / precise — B1 [1]

[Total: 6]

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3	**(a)**	Y-shift **AND** X-shift	B1	[1]
	(b)	connections from either side of cell / resistor to Y inputs	B1	[1]
	(c) **(i)**	2 V cao	B1	[1]
	(ii)	1.4 to 1.6 V	B1	[1]
	(d) **(i)**	dot moves down NOT to centre	B1	[1]
	(ii)	dot moves horizontally / (dot becomes) (horizontal) line (on screen)	B1	[1]
				[Total: 6]
4	**(a)** **(i)**	normal drawn correctly	B1	[1]
	(ii)	*i* marked (ecf incorrectly drawn normal)	B1	[1]
	(iii)	29° to 31° (ecf **(ii)**)	B1	[1]
	(b) **(i)**	line drawn parallel to and 5 cm ±1 mm below upper surface of block	B1	[1]
	(ii)	line through (within 1 mm of) crosses M_1 and M_2 to candidate's line in **(b)(i)**	B1	[1]
	(iii)	ray drawn within block	B1	[1]
	(c)	14° to 20°	B1	[1]
				[Total: 7]

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